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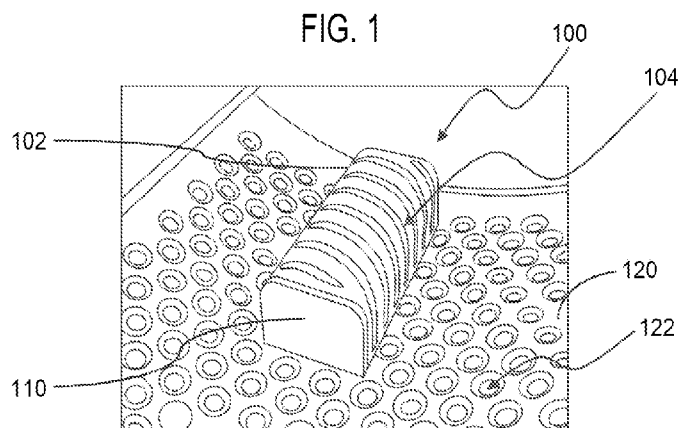
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(54) Title: MICROFIBER FILTER DEVICE FOR LAUNDRY MACHINE



(57) Abstract: A highly-efficient, low-cost microfiber trapping device (100, 600, 700, 800, 900, 1000, 1100) used in consumers' homes to reduce microplastic emissions at the point source- laundry. In an example implementation, the device (100, 600, 700, 800, 900, 1000, 1100) tumbles freely throughout the washer. The device is made of up two reusable, hemispherical halves (602a, 602b) which lock into place and hold a porous, disposable inner filter. In a second implementation, the device is attached to the inside of the machine during the washing cycle. The disposable, porous inner filter, such as one generated via electrospinning, has controllable pore sizes ranging from 10 to 100 μm to trap microfibers in a range of lengths. As the washing machine is run, the device captures microfibers as laundry water flows through it. In another example implementation, the device (100, 600, 700, 800, 900, 1000, 1100) has a rigid, outer shell is a rod-like container of adjustable size with a attachment means (830) for the fixation on the drum of the washing machine.

MICROFIBER FILTER DEVICE FOR LAUNDRY MACHINE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims benefit of priority to U.S. Provisional Patent Application No. 63/505,715 filed June 2, 2023, which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

[0002] Microfibers released from synthetic textiles account for ~35% of the microplastic waste entering the oceans. Millions of microfibers are shed in washing machines, which mechanically and chemically agitate clothing as part of the wash process. Once the microfibers are transported from the home to water treatment plants along with the laundry waste water, they inevitably are released into our ecosystem and make their way into the oceans. Marine organisms ingest these particles, often mistaking them for food. As microplastics have been found to leach plastic additives and adsorb contaminants from their surrounding environments (e.g., heavy metals and polycyclic aromatic hydrocarbons), ingestion of these particles can be toxic. Microplastic ingestion by humans and animals may lead to cell damage and chronic inflammation. It is critical to collect these microfibers before they enter the water waste stream, and as such, collecting microfibers within the washing machine can prevent them from ending up in the oceans.

[0003] There are currently three product categories on the market targeting the collection of microfibers in washing machine wastewater. External filters, like the Filtrol, are installed outside of the washing machine and collect the microfibers before they enter the waste stream. While the Filtrol has high efficiency, the filter is expensive compared to market alternatives and not accessible to everyone due to inconvenient or prohibited installation. Laundry bags are an affordable alternative but have middling efficiency. Laundry bags also have low size specificity and miss smaller fibers. Laundry balls are another affordable alternative, but the products currently on the market also have low efficiency and size specificity. As such, the current competitor landscape is characterized by solutions that can only partially meet the needs of eco-conscious green consumers and there exist glaring gaps in cost-effectiveness, ease of use, and microfiber capture efficiency.

[0004] According to aspects herein, a filter device has a hard, semipermeable outer shell with a fine porous inner core designed to filter microfibers from the inside of the washing machine during a laundry cycle. There are several products on the market designed to eliminate microfibers from the laundry machine source; the filter device as disclosed herein has a combination of features not observed in other conventional products.

SUMMARY OF THE INVENTION

[0005] The invention provides a first-in-class, highly-efficient, low-cost microfiber trapping device used in consumers' homes to reduce microplastic emissions at the point source—laundry. In a first embodiment, the device tumbles freely throughout the washer. The device is made up of two reusable, hemispherical halves which lock into place and hold a porous, disposable inner filter. In a second embodiment, the device is attached to the inside of the machine, during the washing cycle. The outer shell has multiple possible attachment mechanisms, for example, a pin mechanism which interacts with washing machine drainage holes. The disposable, porous inner filter, such as one generated via electrospinning, has controllable pore sizes ranging from 10 to 100 μm to trap microfibers in a range of lengths. As the washing machine is run, the device captures microfibers as laundry water flows through it. Both outer shells are non-permanent additions to the washing machine, acting as universal add-ons to any washing machine. Both also have mechanisms that allow for easy filter removal. The unique combination of outer protective shell and inner sponge-inspired filter generates a novel product designed to trap microfibers during a washing machine cycle.

[0006] According to an aspect of the present disclosure, a device for filtering microfibers from the laundry water during a normal washing cycle is disclosed, the device comprising a hard, protective outer shell and a removable, porous inner filter. The porous inner filter can include an electrospun PLA or cellulose mesh. The porous inner filter can include a PLA or cellulose foam.

[0007] The rigid, outer shell can be made up of two hemispherical halves with varying permeability. The halves can be connected via flexural and snap fit or via a screw or hinge mechanism.

[0008] The rigid, outer shell can be made up of a rod-like container of adjustable size with a method of attachment to the inside of the washing machine. The method of attachment can be generated via tentacles protruding from the base or via a magnet or via a suction cup or via an agitator attachment. The method of attachment can be generated via pins into the washing

machine drainage holes or via hooks into the washing machine drainage holes, or via expandable fit, wedged to the sides of the washing machine drum, or via zip tie loops to some part of the interior of the washing machine.

[0009] According to another aspect of the present disclosure, a device is configured to filter microfibers from laundry water in a washing machine during a washing cycle in which the washing machine has textile items within a drum, the device comprising: a housing having a plurality of openings through which the laundry water is configured to pass, the housing being configured to maintain its integrity in water temperatures between 45°F and 150°F, the housing having an outer section distal from the drum of the washing machine when coupled thereto and an inner section proximate or abutting the drum of the washing machine when coupled thereto, a removable porous, multi-stage filter within the housing, the multi-stage filter including at least two stages, the first stage having a nonwoven material facing the outer section of housing, and the second stage having a material different from the nonwoven material; an attachment feature configured to secure the device relative to an inner surface the drum of the washing machine as the drum is spinning at a rate up to 1000 rotations per minute while exposed to water, the inner surface being below a level of the water, wherein microfibers from the textile items are captured within the second stage of the multi-stage filter as the drum is spinning while exposed to the water.

[0010] The filter can include inner folds or pleats forming a mesh having a gradation of pores whose pore size progressively decreases from an outer portion of the filter toward an interior portion of the multi-stage filter in a second stage thereof to capture the microfibers therein, the microfibers having an average size of 10µm, the plurality of inner folds or pleats including an electrospun sponge-like membrane.

[0011] The first stage can include a nonwoven fine filter wool or fleece material. The material of the second stage can include a mesh.

[0012] The can include a reusable inner core having a base and an elongated member around which the second stage is arranged and abuts the first stage.

[0013] The housing can have a hinge rotatably coupled to a mounting base in which the first stage is inserted. The mounting base can include holes to receive corresponding push rivets to secure the first stage relative to the mounting base.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is an example multi-stage filter device in a closed position and attached to an inner wall of a drum of a washing machine according to an aspect of the present disclosure;

[0015] FIG. 2 shows the filter device of FIG. 1 in an open position to permit visual inspection of the filter state and insertion/removal of new/replacement multi-stage filters.

[0016] FIG. 3A is an exploded view of some parts of the filter device shown in FIG. 1.

[0017] FIG. 3B shows the filter device of FIG. 1 in a closed and open position without the multi-stage filter.

[0018] FIG. 3C shows the filter device of FIG. 1 in an upright position relative to the drum of the washing machine to permit water to drain therefrom due to gravity.

[0019] FIG. 4 illustrates an example replacement or insertion of a mesh filter of the multi-stage filter into a mounting base of the filter device shown in FIG. 1.

[0020] FIG. 5 illustrates an example multi-stage filter according to an aspect of the present disclosure with folds or pleats arranged about its outer surface.

[0021] FIG. 6A illustrates an example filter device having a generally spherical shape with knobs or protrusions arranged about its exterior surface.

[0022] FIG. 6B shows the filter device of FIG. 6A in a closed position retaining a filter or multi-stage filter therein.

[0023] FIG. 7 illustrates an example filter device having a generally spherical shape but lacking the knobs shown in the filter device of FIG. 6A.

[0024] FIG. 8 illustrates a filter device according to an aspect of the present disclosure having an adjustable length to accommodate spacing of holes of a drum of a washing machine.

[0025] FIG. 9 illustrates the filter device shown in FIG. 8 with different attachment mechanisms to secure the filter device to the inner wall of a drum of the washing machine via holes in the drum.

[0026] FIG. 10A is an exploded view of some components of a filter device including a mesh, reusable inner core, and a mounting base that is secured to the drum of a washing machine via its holes using an attachment mechanism having springs and hooks.

[0027] FIG. 10B illustrates the bottom (facing the drum) of the filter device shown in FIG. 10 with the mesh installed into the mounting base.

[0028] FIG. 11A illustrates an example filter device having a generally flattened contour with a cable that is cinched to secure hooks of the filter device to corresponding holes in an inner wall of a drum of a washing machine.

[0029] FIG. 11B illustrates the filter device shown in FIG. 11A with the cable cinched tight to secure the filter device to the drum via hooks.

DETAILED DESCRIPTION

[0030] During a typical washing machine cycle, microplastics can be released from synthetic fabrics such as polyester, nylon, and acrylic. These microplastics are typically shed in the form of fibers due to the mechanical action of washing, the use of detergents, and the friction between clothes. The timing and extent of microplastic release can vary depending on factors such as fabric type, washing machine settings, and detergent composition.

[0031] Research suggests that the majority of microplastics are released from clothing during the washing phase of the cycle, particularly during agitation. The following is a breakdown of the wash cycle and when microplastics are most likely to become intermixed with the water.

[0032] Agitation/Soak: This phase involves the physical agitation of clothes in water, which helps to dislodge dirt, stains, and debris. It is during this stage that significant friction occurs between the fabric fibers, leading to the shedding of microplastics into the water.

[0033] Detergent Dispersion: Detergents contain surfactants and other chemicals that help to lift dirt and grime from clothing and suspend it in the water. The addition of detergent can increase the release of microplastics by further loosening fibers from the fabric.

[0034] Rinsing: In this phase, the washing machine typically drains the soapy water and refills with clean water to rinse away detergent residue and suspended particles. While some microplastics may be flushed out during rinsing, a significant portion may remain suspended in the water due to their buoyancy.

[0035] Spin Cycle: During the spin cycle, the washing machine's drum rapidly rotates to remove excess water from the clothes by centrifugal force. This intense mechanical action can further dislodge microplastics from the fabric and entrain them in the water.

[0036] Drainage: The final phase involves draining the water from the washing machine, carrying any suspended microplastics with it as it exits the machine.

[0037] A consideration in the filter design are the holes in the outer housing that allow the laundry water to pass through the inner filter. In general, the more surface area exposed, the more filtration

possible. Below are considerations given to the various slit geometries, any of which is contemplated herein for use in connection with any filter housing disclosed herein.

[0038] Circular-shaped holes have uniform flow characteristics. They provide a straightforward pathway for fluid flow and are less prone to flow restrictions compared to other shapes. Circular holes can promote efficient particle capture and minimize pressure drop, making them suitable for a wide range of filtration applications.

[0039] Square or rectangular holes offer similar advantages to circular holes in terms of simplicity and ease of fabrication. They provide a uniform flow pathway and can facilitate efficient particle capture. Square or rectangular holes may be preferred in applications where space constraints or specific housing designs necessitate a more geometrically regular shape.

[0040] Triangular Slits: Triangular slits can provide sharp edges that promote particle capture and retention. The triangular shape may also create vortices or eddies in the fluid flow, enhancing particle-surface interactions and filtration efficiency.

[0041] Hexagonal or honeycomb patterns feature interconnected hexagonal-shaped holes arranged in a repeating pattern. These patterns maximize the surface area available for fluid flow while minimizing the amount of material required, resulting in a lightweight and efficient design. Hexagonal or honeycomb patterns can promote uniform flow distribution and enhance particle capture efficiency, making them suitable for high-performance filtration applications.

[0042] Irregular or random hole patterns can offer advantages in certain applications where enhanced fluid mixing or particle-surface interactions are desired. These patterns create diverse flow pathways and flow velocities, which can increase the likelihood of particle capture and retention. However, irregular or random patterns can be more challenging to fabricate and optimize compared to regular geometric shapes.

[0043] While considering water flow, another design constraint for the filters disclosed herein is retaining mechanical durability/strength in the housing shell. Due to the sloshing of water and relatively high centrifugal forces that the filter must endure, while maximizing water flow is important for maximum microparticle capture, the housing must also be able to hold up to water, textiles, and centrifugal forces of the spinning drum of the laundry machine.

[0044] Another consideration is the material or materials of the filters used herein. The following filters and materials can be used in connection with any filter device disclosed herein. An inspiration for the filters herein is sea sponges, which are nature's filters and filter feeders. Every day, a sea sponge can filter between two and 20 cubic meters of water per kilogram of bodyweight.

[0045] Unlike a soft, squishy kitchen sponge, sea sponges are stiff and strong. They have an amazingly complex skeleton that consists of a spiral hierarchical design of fibers, known as spicules, no larger than a human hair. This intricate multi-layered and cylindrical shape allows water to flow in and out of its body while enduring a high-pressure environment.

[0046] Taking cues from sea sponges, the bio-inspired sponge-like membrane filter herein provides unparalleled convenience and efficiency. Through physical entrapment and adsorption mechanisms, high porosity, high surface area, and open-celled structures, the filter herein achieves targeted, reliable, and effective microplastics capture during the laundry cycle.

[0047] Similar to how the sea sponge's unique structural pattern is composed of different thicknesses and angles of orientation at specific distances, the inner filter according to some aspects herein is spun to have a gradation of pores (as you go from the outside closer to the core, each layer is a more porous, tighter mesh).

[0048] Another natural inspiration comes from suspension feeders like oysters, which use their gills to retain suspended particles while cilia actively pump water through small openings. This concept influenced the filter's design within the washing machine drum. By positioning the filter inside the drum, the natural flow of water and the centrifugal force of the spin cycle are harnessed, eliminating the need for external pumps, batteries, or additional energy usage. The devices herein can incorporate inner channels and doors that open and close depending on water flow direction, mimicking the behavior of cilia to further enhance filtration efficiency.

[0049] The filter characteristics or attributes considered herein include geometries, densities, porosities, thicknesses, composition, form factors, and surface modifications. The filter must be suited for the specific high-flux, basic (7-10 pH), temperature range (60°F - 130°F or 45°F - 150°F), and turbulent environment inside the spinning or agitating drum of the washing machine. Examples of materials for the filter include: Cellulose acetate/acetone electrospun membrane, cellulose acetate/acetone aerogels, open-cell polyurethane-based foam (35 & 45 PPI), non-woven fine filter wool, polyurethane sponge, cellulose sponge, silicone sponge, pleated cellulose membrane filter (3, 8, 11, 20-25µm), unpleated cellulose membrane filter (3, 8, 11, 20-25µm), pleated coarse apertured 100% viscose spunlace nonwoven, unpleated coarse apertured 100% viscose spunlace nonwoven, and stainless steel mesh (100 & 100 micron).

[0050] Suitable surface modifications/adsorbents include: Fe₃O₄ + PDMS coating, which repels water while attracting nonpolar plastics (i.e., polypropylene + polyethylene), activated carbon to

increase surface area, and amine-functionalized: precedent shown to remove lead and heavy metals from water.

[0051] Surprising results from testing the influence of orientation, shape, length, thickness, and porosity on capture efficiency in real-world environment include the following observations: 30 PPI vs 45 PPI only marginally different in pre-milled microfiber capture efficiency; aerogels created by treating 2D cellulose acetate/acetone electrospun mats with sodium borohydride and vacuum drying exhibit similar capture efficiency, but are extremely fragile and exhibited poor mechanical strength; better performing trials favored 4.5" length filters over 9" length; most of the trials that worked well only had 2 layers in thickness; in most trials, the inside of the filter (facing the drum of the wall) exhibits higher visual indication of capture than the outside; areas of filter medium compressed by elastic band or folded exhibit increased visual indication; gradation, i.e., layering non-woven fine filter wool with stainless steel mesh plate (installed flush against surface of the drum) helps reduce microfibers lost due to high centrifugal force during the rinse and spin cycles; lifespan tests indicate that the filters can last at least 10 cycles without significant resistance; however, there is potential in back-flushing during the wash cycle. The microfibers are captured on the surface of the pleated filter in a moderately reversible way, and aspects of the present disclosure leverage the centrifugal force and a pulse to knock them off again and move the microfibers to a separate compartment where they get stored. Filters could maintain longer life over longer periods of time, preventing clogging and fouling.

[0052] Gradient-layered filters of specific mesh sizes (macroparticles [250 μm], human hair [100 μm], pet hair, [50 μm], microplastics [10 μm]) are also contemplated.

[0053] Some aspects include a multi-stage filter having at least an inner filter and a second stage filter. The inner filter is designed with a large surface area to maximize contact with the water flow. This can be achieved, for example, by using a filter material that can be folded or pleated to increase the effective surface area without significantly increasing the size of the filter housing.

[0054] The filter device is installed in a location where the water most often passes. Horizontal and vertical orientations of the filter are contemplated; however, it has been found that horizontally orienting the filter at the bottom of a top loader and as far as back as possible in a front loader produces optimum filtration.

[0055] The filter design directs the flow of water to the filter media by spanning the width of the drum and using walls to guide the water into the media.

[0056] The filter retains the microplastics from the fast-spinning and agitation forces during its second stage of filtration via its layered design. The nonwoven mat captures the microfibers during the initial sloshing of water in the machine. After this capture, the microfibers are pushed out of the nonwoven mat and onto a fine mesh that holds and retains the fibers.

[0057] Via gradation/layered membrane filter design: Implementing a multi-stage filtration system where water passes through multiple layers of filters with progressively smaller pore sizes can effectively capture microplastics. This approach distributes the force exerted by the spinning drum across multiple filters.

[0058] FIG. 1 illustrates a filter device 100 having an elongated shape and generally flat profile with a contoured shape that matches the style of an agitator inside the drum of a washing machine. An outer shell or housing 102 has a hard shell with a filleted rectangular shape that hugs the drum, reduces the chance that clothing, towels, bedding, or other textiles will get stuck or trapped by the housing 102. A reusable inner core 320 (FIG. 3A) inspired by a popsicle stick mold allows for easy loading and unloading of the filter without having to touch the inner filter 210 (FIG. 2). A self-drainage design flips upwards with a hinge to use gravity when not in use to drain away water, which can be best seen in FIG. 3B. A mesh filter 220 can be easily loaded into the housing 102 by, for example, push rivets that are pressed into corresponding mounting holes 330a (FIG. 3C), which are hidden in the mounting base 310. FIG. 4 illustrates how the mesh filter 220 is inserted into the mounting base 310 by press-fitting the push rivets into place into their corresponding mounting holes 330a. By flipping the device up in the orientation shown in FIG. 3C, visual inspection of the filter is possible while also allowing water to drain away from the device 100.

[0059] The device 100 is configured to filter microfibers from laundry water in a washing machine during a washing cycle in which the washing machine has textile items within a drum. Textile items can include any one or more of clothing, bedding, towels, and the like that are washed with a washing/laundry machine that can have a top-loading or front-loading configuration. The filter devices herein can be used in either configuration.

[0060] The housing 102 has openings through which the laundry water is configured to pass when the drum is filled with water during a wash or rinse cycle. The housing 102 is configured to maintain its integrity in water temperatures between 45°F and 150°F, or any temperature range of water present in a washing machine. The housing 102 has an outer section distal from the drum 120 of the washing machine when coupled thereto and an inner section (see mounting base 310 in FIG. 3B) proximate or abutting the drum 120 of the washing machine when coupled thereto.

[0061] A removable porous, multi-stage filter 210, 220 is present within the housing, the multi-stage filter including at least two stages, the first stage having a nonwoven material facing the outer section of housing, and the second stage 210 has a material different from the nonwoven material 220.

[0062] An attachment feature (see, e.g., FIG. 9) is configured to secure the device 100 relative to an inner surface the drum 120 of the washing machine as the drum 120 is spinning at a rate up to 1000 rotations per minute while exposed to water having a temperature between 60°F and 130°F or between 45°F and 150°F, the inner surface being below a level of the water.

[0063] The microfibers from the textile items are captured within the second stage 210 of the multi-stage filter 210, 220 as the drum 120 is spinning while exposed to the water.

[0064] The filter can include inner folds or pleats 504 (FIG. 5) forming a mesh having a gradation of pores whose pore size progressively decreases from an outer portion of the filter 500 toward an interior portion 510 of the multi-stage filter 100 in a second stage thereof to capture the microfibers therein. The microfibers can have an average or minimum captured size of 10µm. The inner folds or pleats 504 can include an electrospun sponge-like membrane. While the example shown in FIG. 5 is a cylindrical shaped filter, in other aspects, the filter can be shaped as an elongated rectangle 210, such as seen in FIG. 3A. The filter housing 502 retains the folds 504 and inner portion 510.

[0065] The first stage 220 can include a nonwoven fine filter wool or fleece material, and the material of the second stage 210 can include a mesh.

[0066] A reusable inner core 212 (see FIG. 3A) has a base 322 and an elongated member (rod-shaped in the example illustration) 320 around which the second stage 210 is arranged and abuts the first stage 220 when the mounting base 310 is closed.

[0067] The housing can have a hinge 330 rotatably coupled to the mounting base 310 in which the first stage 220 is inserted.

[0068] The mounting base can include holes 330a,b,c (FIG. 3C) to receive corresponding push rivets to secure the first stage 220 relative to the mounting base 310 (FIG. 4).

[0069] FIG. 5 as discussed above shows an example filter 500 representing a pleated surface filter with a gradation of pores with larger pores on the outside and increasingly smaller pores towards the center. The pleated surface filter 500 is designed to maximize surface area and trap the most particles possible, while implementing a multi-stage filtration system where water passes through multiple layers of filters with progressively smaller pore sizes distributes the force exerted by the

spinning drum 120 across multiple filters. The filter 500 can include a backwash compartment to trap backwashed particles.

[0070] In FIG. 6A, a microfiber filter ball device 600 having two separated hemispherical halves 602a, 602b are portrayed and at least one half includes a porous inner filter. Knobs or protrusions 610 are shown as their use aid in preventing the ball 600 from getting trapped in the clothing. One possible mechanism of adhesion between the layers, the snap mechanism, is shown in this illustration. The snap mechanism includes a tongue-and-groove design in which a tongue 604b, 606a fits into a corresponding groove 604a, 606b on the respective hemispherical halves 602a, 602b. Slots or slits of varying sizes and shapes 612 are illustrated to permit laundry water to pass through and to entrap microplastics and other microparticles within the porous inner filter.

[0071] The filter 600 has a semipermeable, hard outer shell 602a, 602b, varying in terms of slit size 612 and quantity of holes 612 to optimize water flow through the device 600. The shell 602, both with and without outer knobs 610, is spherical in shape when the two hemispheres 602a, 602b of the filter 600, made of a material such as thermoplastic polyurethane, are screwed, hinged, or snapped together. Within the hard shell 602 is placed a porous, absorbent inner filter such as resembling a sea sponge or any other filter disclosed herein. The material can be manufactured with pores of varying sizes on the micrometer scale to ensure capture of different lengths of microfibers including microplastics. The inner filter is porous and can be made by techniques such as electrospinning or foaming and with materials such as PLA and/or cellulose. The inner filter is nestled between the two hemispheres 602a, 602b of the outer shell when used. Together, the parts 602a, 602b of the device connect together and can be tossed into the laundry with the clothes during a laundry wash cycle.

[0072] In FIG. 6B, the first embodiment is shown when the two hemispheres 602a, 602b are snapped together using the snap mechanism to form a closed ball 600 that can be tossed into a drum of a laundry machine along with clothes and other textile items. The knobs 610 allow the ball 600 to get caught up in the textile items in the drum of the laundry machine as the ball 600 tumbles throughout the textile items so the ball's movement path is random and allows other textile items to "grab" onto one of the knobs 610 to maximize the ball's traversal through the water of the drum to capture microparticles along its journey.

[0073] In FIG. 7, a ball 700 is shown with increased slit size 712 and a lack of knobs that are shown on the ball 600. The slits 712 are elongated and resemble slices cut out of a lemon wedge and taper as they extend from the larger diameter center of the ball 700 toward the poles

of the ball 700. This ball 700 will allow increased water flow compared to the ball 600 and its relatively smooth exterior surface allows the ball 700 to travel freely amongst the textiles in the drum of the laundry machine relatively unimpeded.

[0074] In FIG. 8, a rendering of another filter 800 is shown having a rod-like shape. The adjustable feet 814a, 814b with pin holes 816, 818 are shown. Rivets 822, 824 are pushed through the pin holes 816, 818 to secure the filter 800 to the inner drum of the laundry machine. Rivets 830 of different sizes can be included with a kit including the filter 800 to allow the filter 800 to be secured to drum holes of different sizes. The screw mechanism 820 demonstrates adjustable length along the body of the filter 800. The length of the screw mechanism 820 can be adjusted along the direction of arrow A to align the rivets 822, 824, 830 with the holes of the drum of the laundry machine. An inner filter such as any disclosed herein is contained inside the filter 800.

[0075] The device 800 is designed such that it attaches to the inside (drum) of the washing machine. It is a rod-shaped container with adjustable feet and holes available for pins. Its length can be modified via a screw mechanism along its body and water flows through via slits 840 along the shell's outer surface. The inner filter can vary in geometry from the device 600, 700 but can include the same material. The attachment mechanism is designed to be removable and reusable over repeated laundry cycles.

[0076] FIG. 9 illustrates an example of a filter device 900 that is removably attached to a wall of a drum of a laundry machine, like the device 800 shown in FIG. 8. Different attachment mechanisms are shown, including flexible hooks, concealed hooks that hook into the holes 122 (shown in FIG. 1) of the drum, push-in rivets that are pressed into holes of the drum as shown in FIG. 8, cross-shaped slots that receive a corresponding cross-shaped plug that connects to holes of the inner wall of a drum, and concealed magnets that magnetically secure the filter device 900 to a magnetized inner wall of the drum. Most washing machine drums are magnetic, using 400 series ferritic stainless steel. A lower-profile housing (e.g., the device 1100 shown in FIG. 11A) can be used when the attachment mechanism is magnets to reduce a height of the filter and reduce the chances of its getting knocked off the drum.

[0077] FIG. 10 illustrates another filter device 1000 having a drum baffle 1002 with a lifter shape, a reusable inner core 1004, and a mesh panel 1006 with a filter membrane. The device 1000 includes springs and hook attachments 1010, which are configured to removably secure the device 1000 to holes inside drums of a laundry machine. In this exploded view shown in

FIG. 10A, the mesh panel together with the filter membrane 1006 slot together inside of the baffle housing (not shown, but an example housing 102 can be seen in FIG. 3A). FIG. 10B illustrates an inside view of the assembled baffle and filter.

[0078] FIG. 11A illustrates another example of a filter device 1100 featuring a soft-shell housing 1102 made of a soft shell material, and a hook attachment 1104 cinched together with elastic cable 1106 and lace lock 1108. The cable 1106 is pulled taut along the direction of arrow C shown in FIG. 11B, and then locked in place by the lace lock 1108 to cinch all of the hook attachments 1104 (six shown in the example of FIG. 11A) in a secure relationship relative to an inner wall of a drum of a laundry machine.

[0079] While the invention has been described with reference to the embodiments above, a person of ordinary skill in the art would understand that various changes or modifications may be made thereto without departing from the scope of the claims.

CLAIMS

1. A device for filtering microfibers from the laundry water during a normal washing cycle, said device comprising a hard, protective outer shell and a removable, porous inner filter.
2. The device defined in claim 1, in which the porous inner filter comprises an electrospun PLA or cellulose mesh.
3. The device defined in claim 1, in which the porous inner filter comprises a PLA or cellulose foam.
4. The device defined in claim 1, in which the rigid, outer shell is made up of two hemispherical halves with varying permeability.
5. The device defined in claim 4, in which the halves are connected via flexural and snap fit.
6. The device defined in claim 4, in which the halves are connected via a screw mechanism.
7. The device defined in claim 4, in which the halves are connected via a hinge mechanism.
8. The device defined in claim 1, in which the rigid, outer shell is made up of a rod-like container of adjustable size with a method of attachment to the inside of the washing machine.
9. The device defined in claim 8, in which the method of attachment is generated via tentacles protruding from the base.
10. The device defined in claim 8, in which the method of attachment is generated via a magnet.
11. The device defined in claim 8, in which the method of attachment is generated via a suction cup.
12. The device defined in claim 8, in which the method of attachment is generated via an agitator attachment.

13. The device defined in claim 8 in which the method of attachment is generated via pins into the washing machine drainage holes.
14. The device defined in claim 8 in which the method of attachment is generated via hooks into the washing machine drainage holes.
15. The device defined in claim 8 in which the method of attachment is generated via expandable fit, wedged to the sides of the washing machine drum.
16. The device defined in claim 8 in which the method of attachment is generated via zip tie loops to some part of the interior of the washing machine.
17. A device configured to filter microfibers from laundry water in a washing machine during a washing cycle in which the washing machine has textile items within a drum, the device comprising:
 - a housing having a plurality of openings through which the laundry water is configured to pass, the housing being configured to maintain its integrity in water temperatures between 45°F and 150°F, the housing having an outer section distal from the drum of the washing machine when coupled thereto and an inner section proximate or abutting the drum of the washing machine when coupled thereto,
 - a removable porous, multi-stage filter within the housing, the multi-stage filter including at least two stages, the first stage having a nonwoven material facing the outer section of housing, and the second stage having a material different from the nonwoven material;
 - an attachment feature configured to secure the device relative to an inner surface the drum of the washing machine as the drum is spinning at a rate up to 1000 rotations per minute while exposed to water, the inner surface being below a level of the water,wherein microfibers from the textile items are captured within the second stage of the multi-stage filter as the drum is spinning while exposed to the water.

18. The device of claim 17, wherein the filter includes a plurality of inner folds or pleats forming a mesh having a gradation of pores whose pore size progressively decreases from an outer portion of the filter toward an interior portion of the multi-stage filter in a second stage thereof to capture the microfibers therein, the microfibers having an average size of 10 μ m, the plurality of inner folds or pleats including an electrospun sponge-like membrane.
19. The device of claim 17, wherein the first stage includes a nonwoven fine filter wool or fleece material.
20. The device of claim 17, wherein the material of the second stage includes a mesh.
21. The device of claim 17, further comprising a reusable inner core having a base and an elongated member around which the second stage is arranged and abuts the first stage.
22. The device of claim 17, wherein the housing has a hinge rotatably coupled to a mounting base in which the first stage is inserted.
23. The device of claim 23, wherein the mounting base includes a plurality of holes to receive corresponding push rivets to secure the first stage relative to the mounting base.

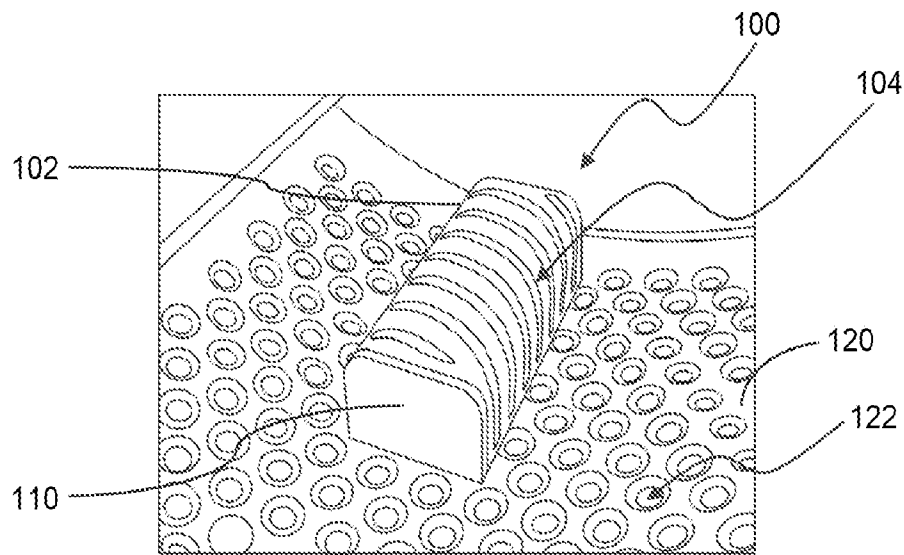


FIG. 1

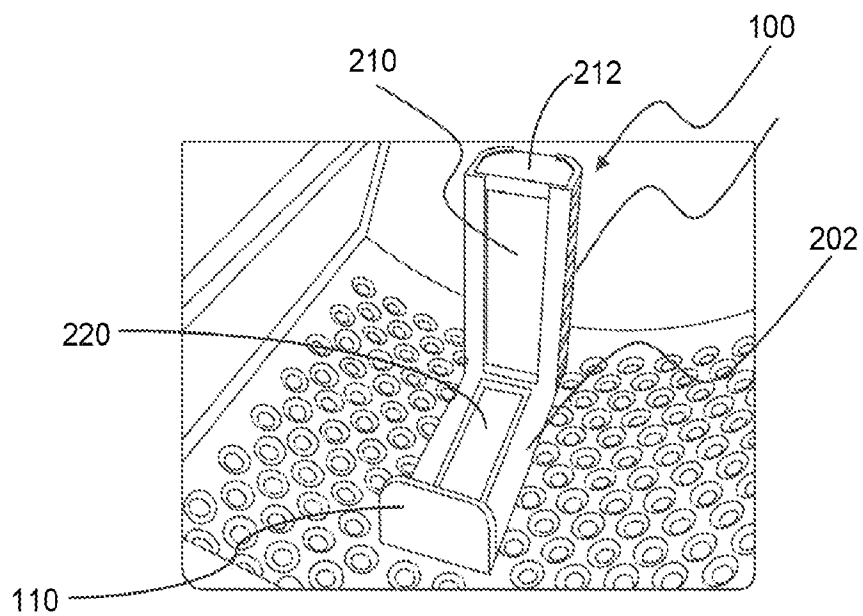


FIG. 2

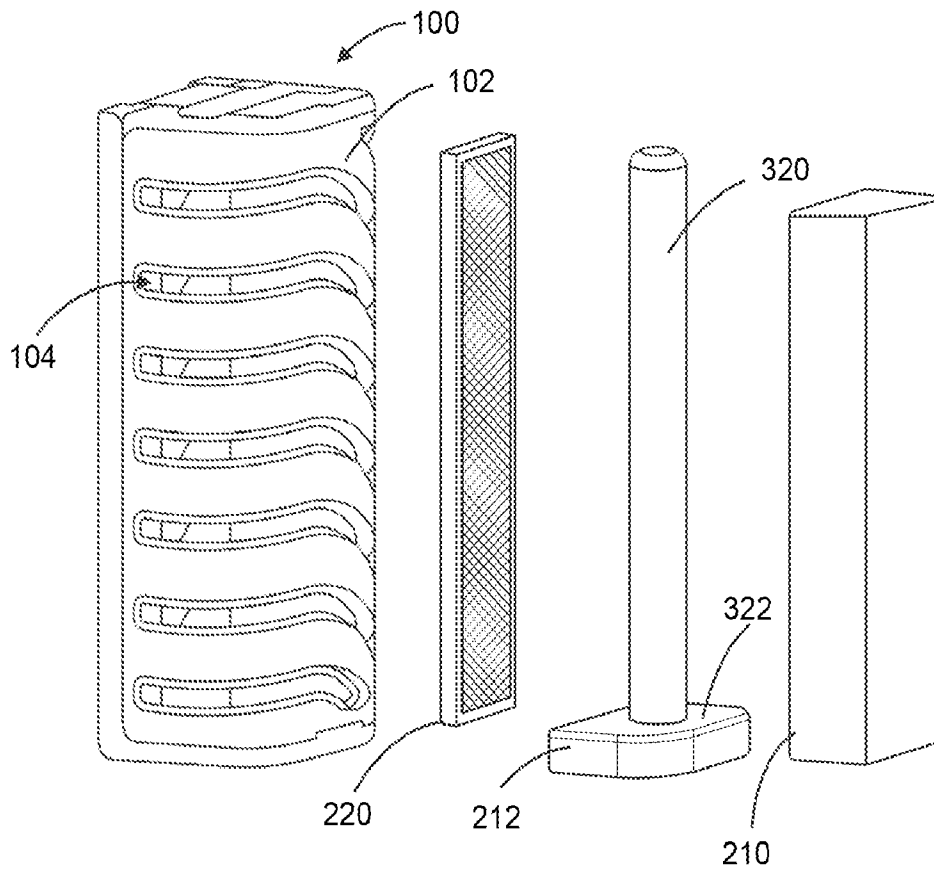


FIG. 3A

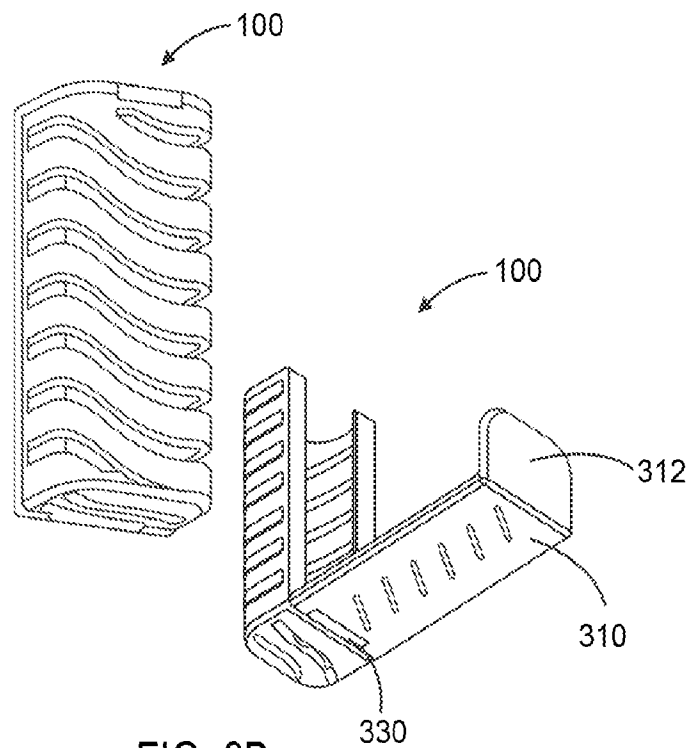


FIG. 3B

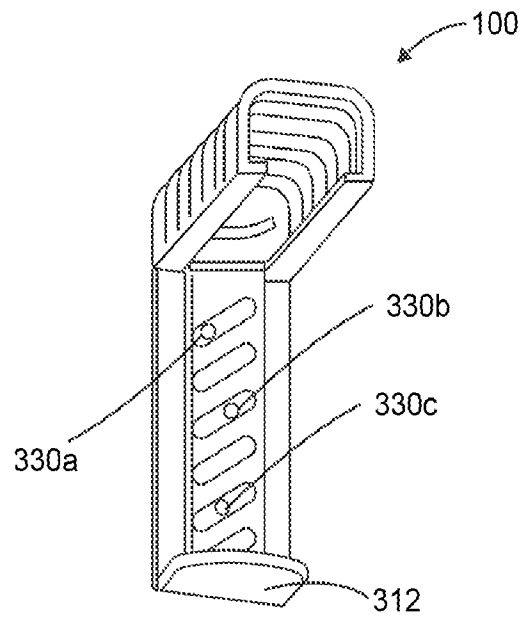


FIG. 3C

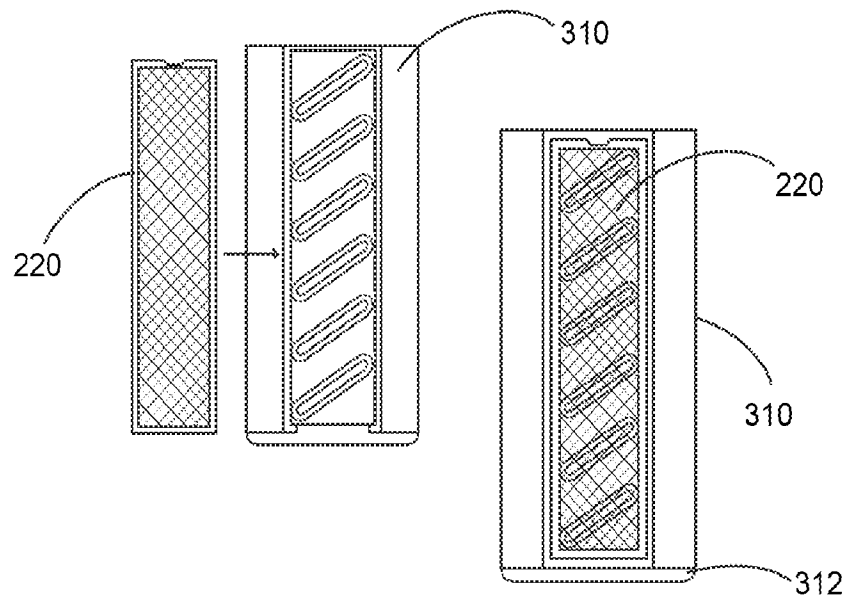


FIG. 4

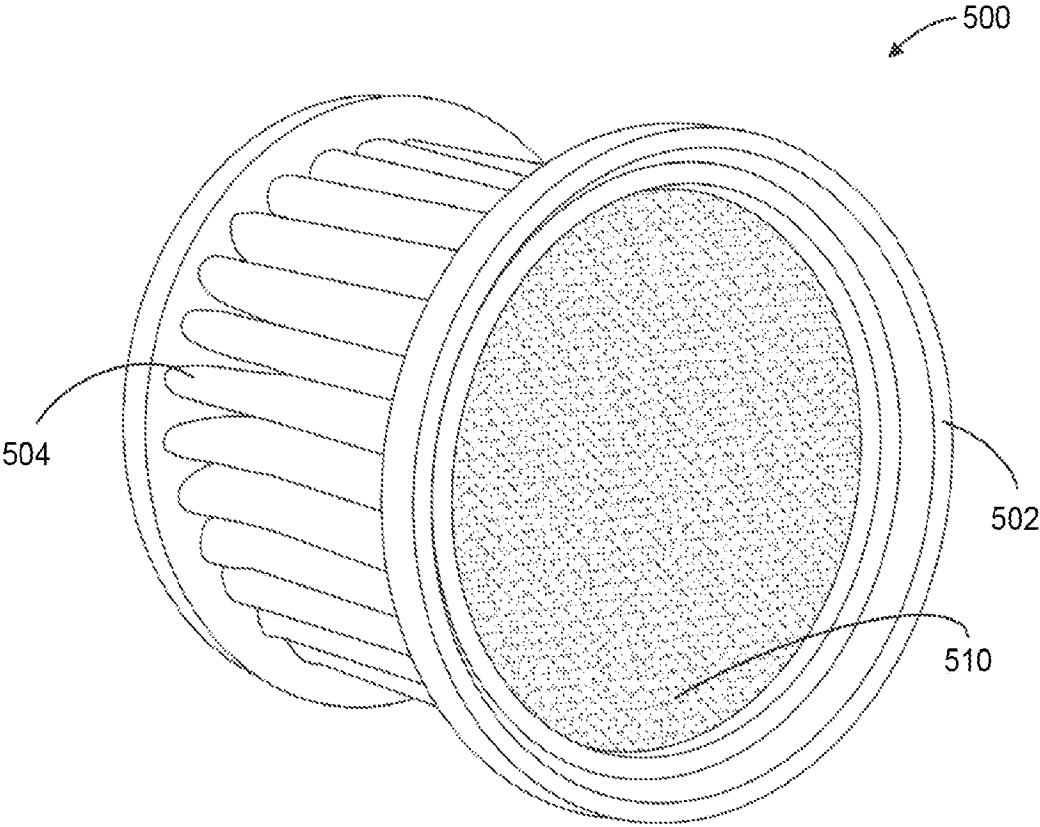


FIG. 5

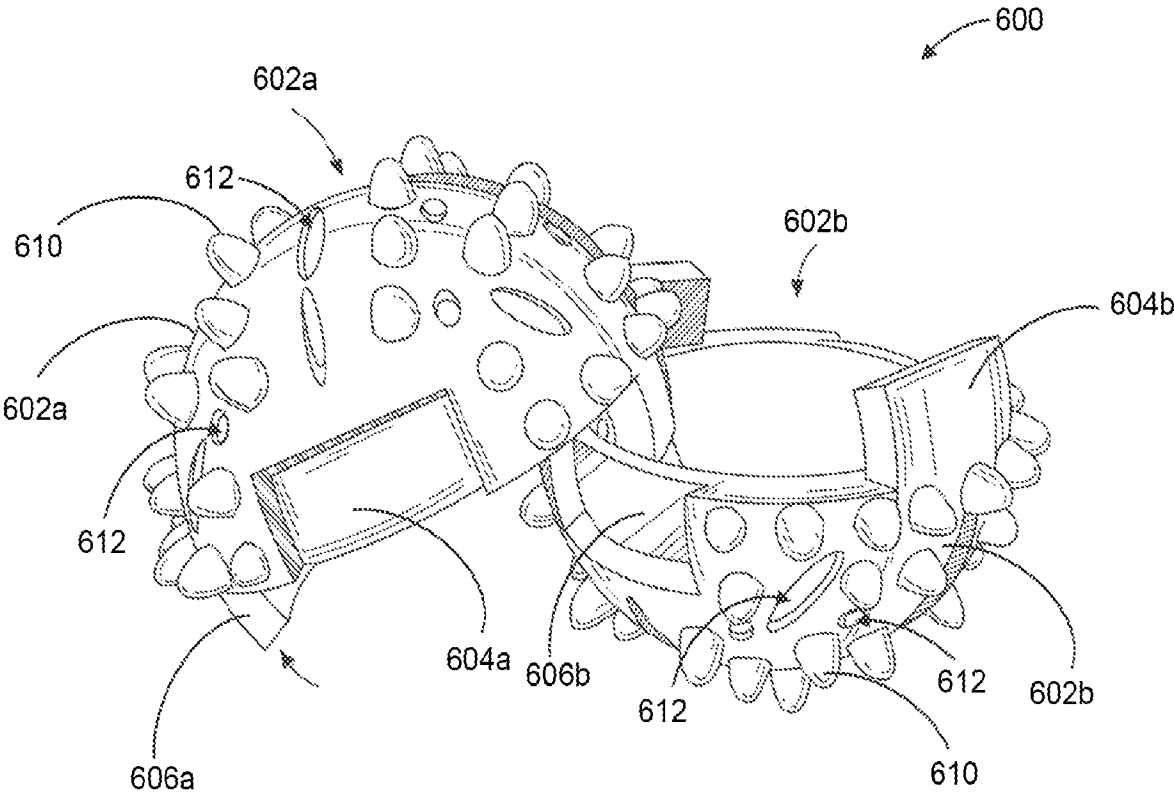


FIG. 6A

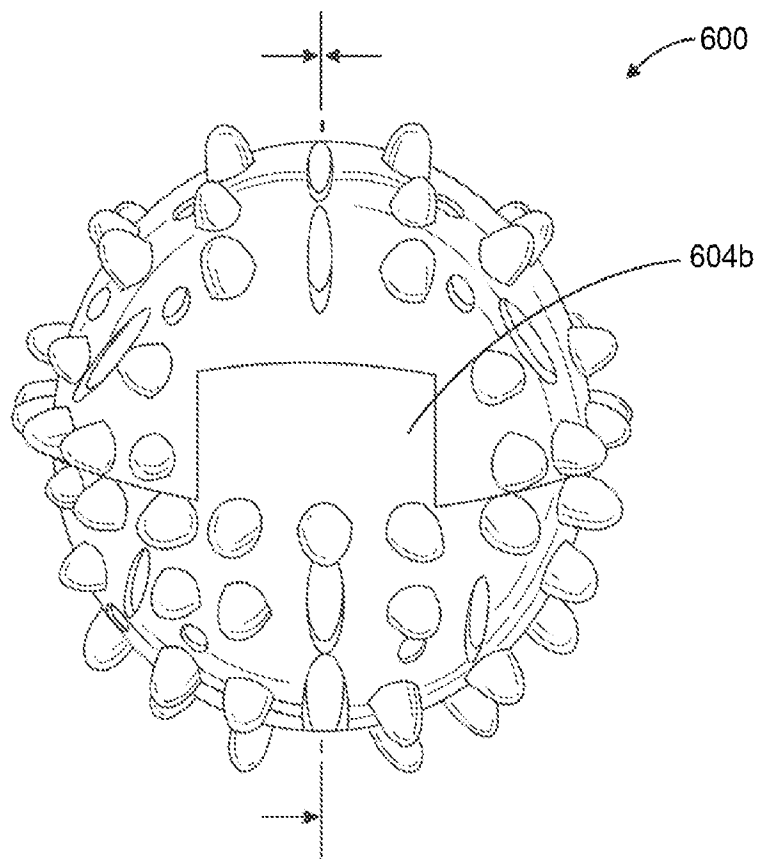


FIG. 6B

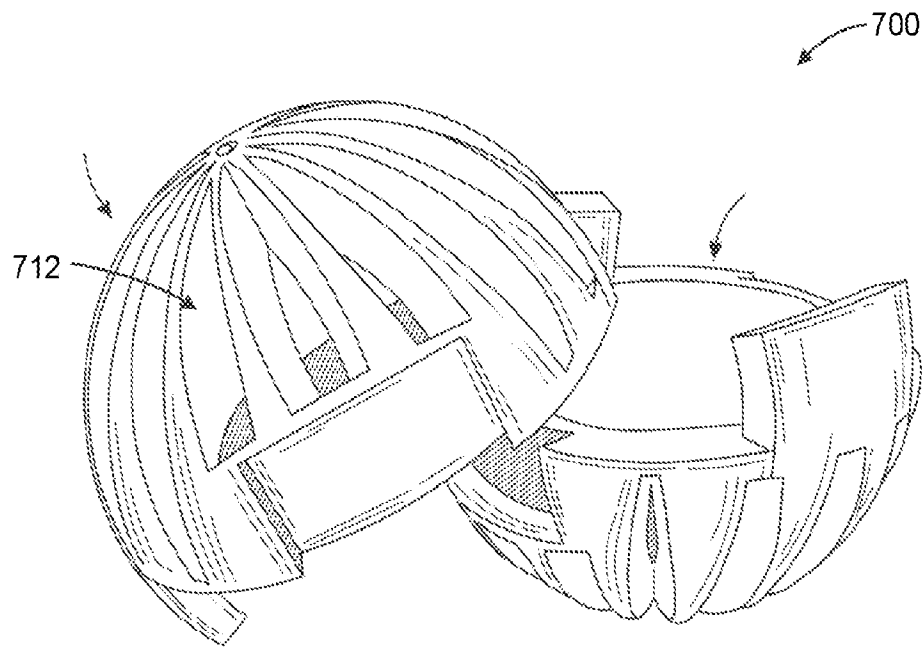


FIG. 7

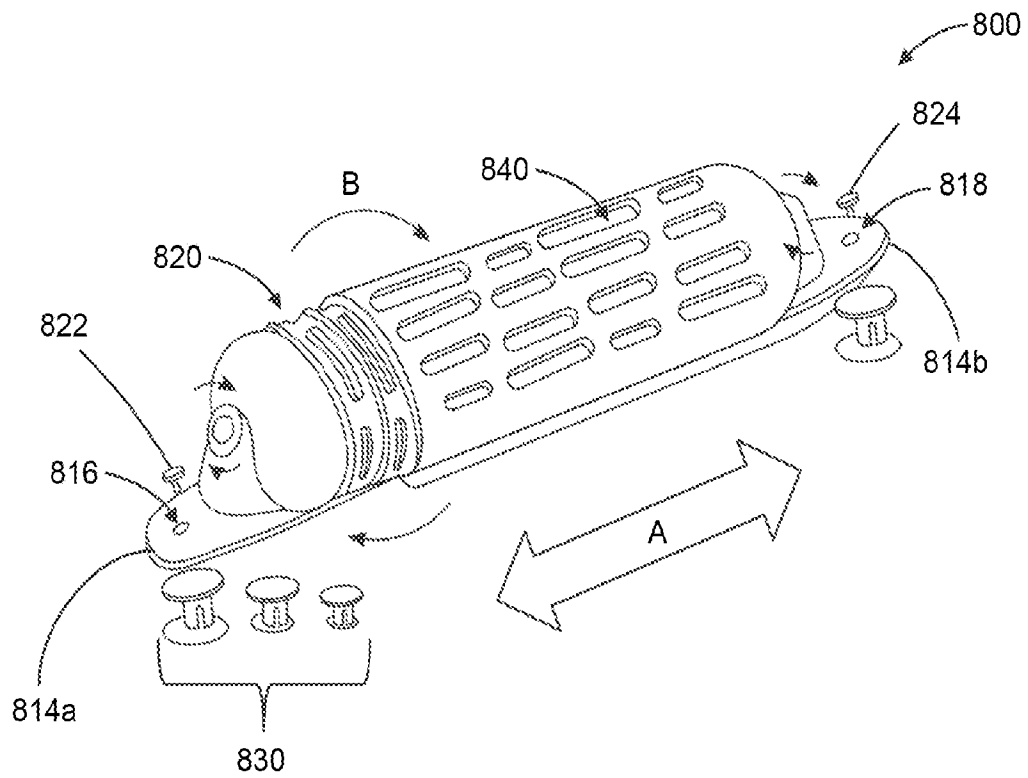


FIG. 8

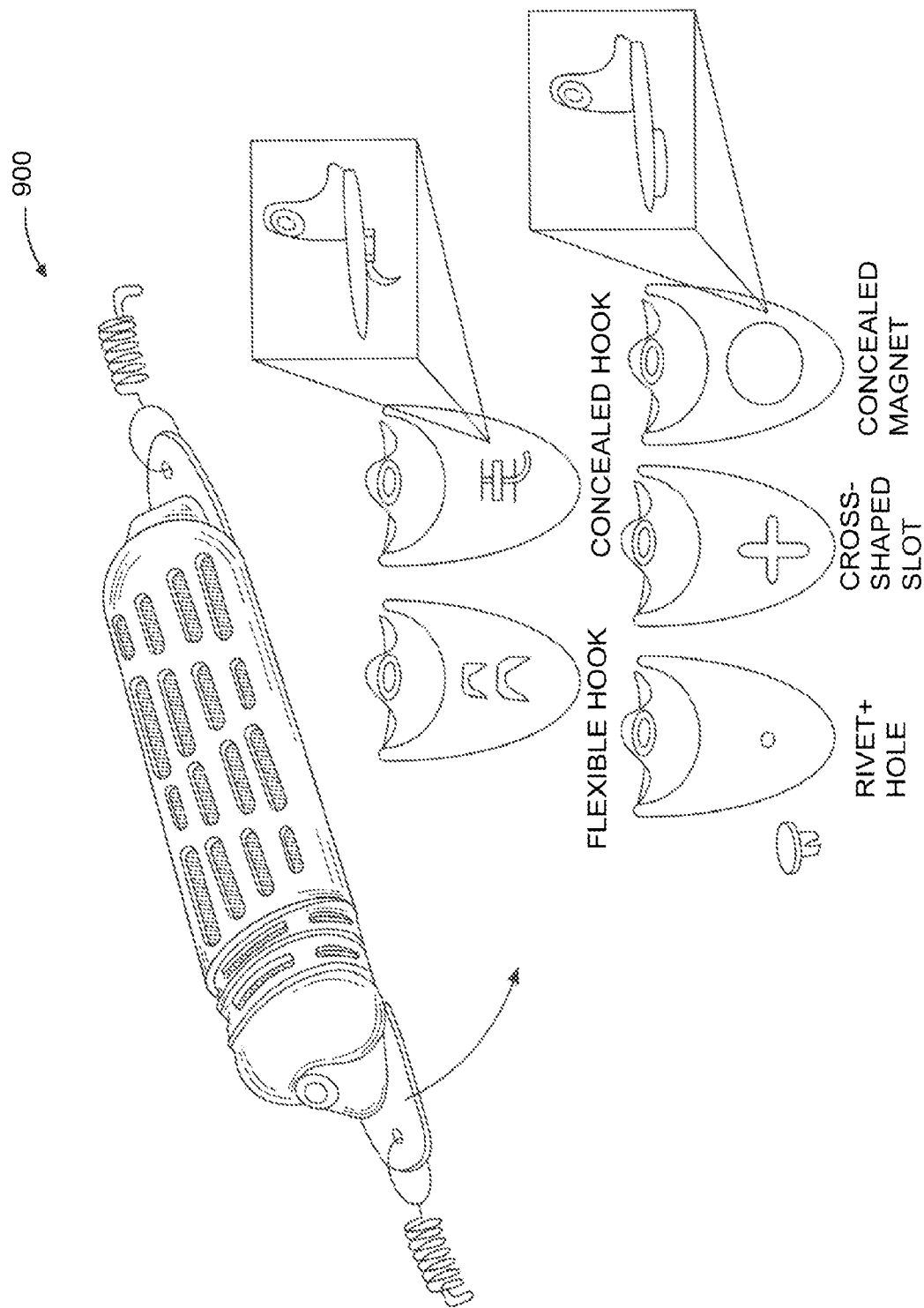


FIG. 9

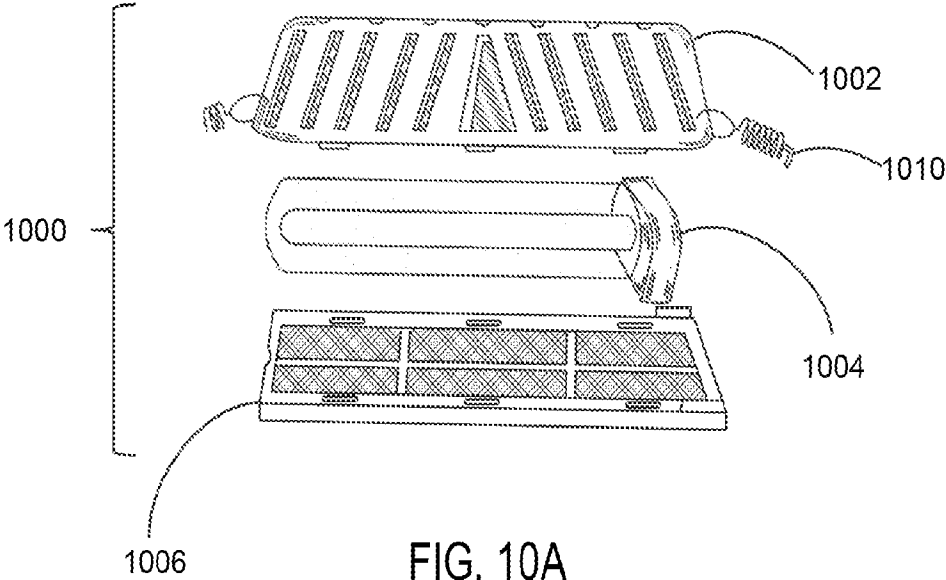


FIG. 10A

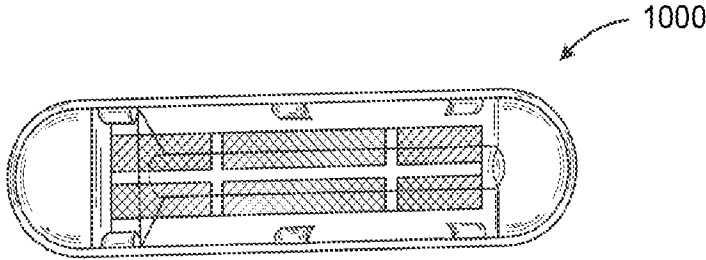


FIG. 10B

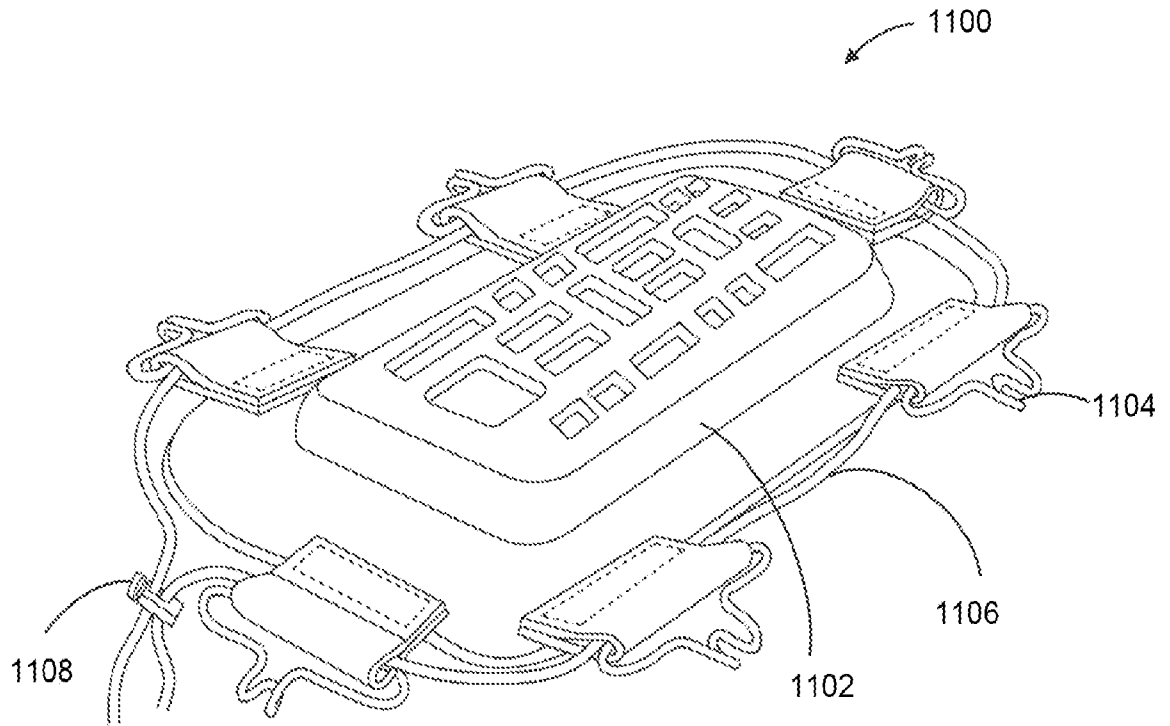


FIG. 11A

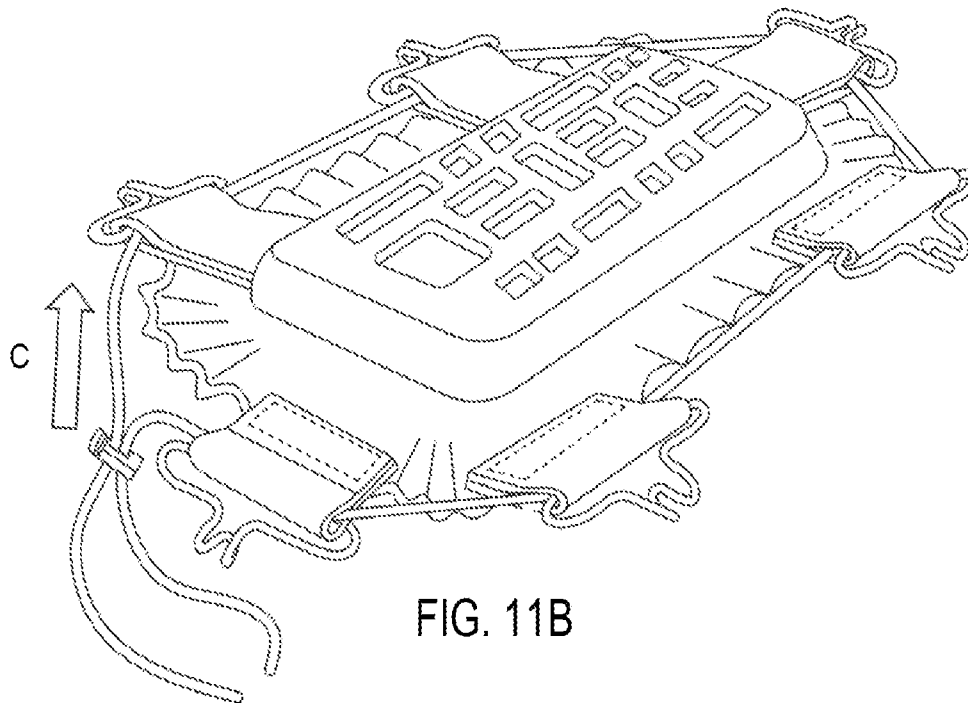


FIG. 11B

INTERNATIONAL SEARCH REPORT

International application No

PCT/US2024/032274

A. CLASSIFICATION OF SUBJECT MATTER

INV. D06F39/10

ADD. B01D39/16 B01D39/18

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

D06F B01D

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO 2022/075941 A1 (ARCELIK AS [TR]) 14 April 2022 (2022-04-14)	1-7
A	page 4, line 24 - page 8, line 10; figures -----	8-23
X	KR 102 374 000 B1 (JANG JUN YEONG [KR]) 14 March 2022 (2022-03-14)	1-7, 17-23
A	paragraph [0012] - paragraph [0056]; figures -----	8-16
A	US 2023/016651 A1 (DI MAGGIO ANTONIO [DE] ET AL) 19 January 2023 (2023-01-19) paragraph [0050] - paragraph [0072] -----	2,3
A	WO 2020/174951 A1 (FUJIFILM CORP [JP]) 3 September 2020 (2020-09-03) paragraph [0012] - paragraph [0069]; figures ----- -/-	2,3

☒ Further documents are listed in the continuation of Box C.

☒ See patent family annex.

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"O" document referring to an oral disclosure, use, exhibition or other means

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"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

23 July 2024

Date of mailing of the international search report

20/08/2024

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INTERNATIONAL SEARCH REPORT

International application No

PCT/US2024/032274

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	US 2018/319146 A1 (HWANG JUN SIK [KR] ET AL) 8 November 2018 (2018-11-08) paragraph [0046] - paragraph [0105]; figures -----	2,3
X	KR 102 172 113 B1 (KOREA INST IND TECH [KR]) 30 October 2020 (2020-10-30) paragraph [0014] - paragraph [0004]; figures -----	1,4-7
A	WO 2020/216890 A1 (TECHNOTRADE VERWALTUNGSGESELLSCHAFT MBH [DE]) 29 October 2020 (2020-10-29) page 5, line 1 - page 8, line 11; figures -----	2,3,8-23
A		1,4-7

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/US2024/032274

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
WO 2022075941 A1	14-04-2022	EP 4225988 A1 TR 202015872 A1 WO 2022075941 A1	16-08-2023 21-04-2022 14-04-2022

KR 102374000 B1	14-03-2022	NONE	

US 2023016651 A1	19-01-2023	CN 115608019 A DE 102021207442 A1 EP 4119212 A1 KR 20230011240 A US 2023016651 A1	17-01-2023 19-01-2023 18-01-2023 20-01-2023 19-01-2023

WO 2020174951 A1	03-09-2020	CN 113453780 A JP WO2020174951 A1 TW 202100221 A US 2021387123 A1 WO 2020174951 A1	28-09-2021 23-12-2021 01-01-2021 16-12-2021 03-09-2020

US 2018319146 A1	08-11-2018	BR 112015019842 A2 CN 104994928 A KR 20140103866 A US 2015360157 A1 US 2018319146 A1 US 2020094539 A1	18-07-2017 21-10-2015 27-08-2014 17-12-2015 08-11-2018 26-03-2020

KR 102172113 B1	30-10-2020	NONE	

WO 2020216890 A1	29-10-2020	DE 202019102297 U1 EP 3959370 A1 WO 2020216890 A1	27-07-2020 02-03-2022 29-10-2020
